Unit outline_

COMP5318: Machine Learning and Data Mining

Semester 2, 2026 [Normal evening] - Camperdown/Darlington, Sydney

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Overview

Machine learning is the process of automatically building mathematical models that explain and generalise datasets. It integrates elements of statistics and algorithm development into the same discipline. Data mining is a discipline within knowledge discovery that seeks to facilitate the exploration and analysis of large quantities for data, by automatic and semiautomatic means. This subject provides a practical and technical introduction to machine learning and data mining. Topics to be covered include problems of discovering patterns in the data, classification, regression, feature extraction and data visualisation. Also covered are analysis, comparison and usage of various types of machine learning techniques and statistical techniques.

Unit details and rules

Academic unit	Computer Science
Credit points	6
Prerequisites ?	None
Corequisites ?	None
Prohibitions ?	COMP4318 or OCMP5318
Assumed knowledge ?	Experience with programming and data structures as covered in COMP2123 or COMP2823 or COMP9123 (or equivalent unit of study from different institutions). Discrete mathematics and probability (e.g. MATH1064 or equivalent); linear algebra and calculus (e.g. MATH1061 or equivalent)
Available to study abroad and exchange students	Yes

Teaching staff

Coordinator	William Zhi, weiming.zhi@sydney.edu.au
Lecturer(s)	Nguyen Tran, nguyen.tran@sydney.edu.au

Assessment

ⓘ The census date for this unit availability is **31 August 2026**

Details | Criteria

Type	Description	Weight	Due	Length	Use of AI
Written exam 🚧	**Final exam** Exam	60%	Formal exam period	2 hours	AI prohibited
Outcomes assessed: LO1 LO3 LO4					
Out-of-class quiz	**Weekly homeworks** Homework exercises	5%	Multiple weeks	-	AI allowed
Outcomes assessed: LO1 LO2 LO4 LO5					
Practical skill 👥	**Assignment 1** Computer program and report. Group assignment.	15%	Week 07 **Due date**: 18 Sep 2026 at 23:59	-	AI allowed
Outcomes assessed: LO1 LO2 LO3 LO4 LO5					
Creative work 👥	**Assignment 2** Computer program, report and presentation. Group assignment.	20%	Week 12 **Due date**: 30 Oct 2026 at 23:59	-	AI allowed
Outcomes assessed: LO1 LO2 LO3 LO4 LO5					

🚧 = hurdle task ? 👥 = group assignment ?

Assessment summary

- Weekly homeworks
- Oral interviews during tutorial sessions in Weeks 3, 5, 7, 9, and 11
- Assignment 1 – writing a computer program to solve a given task and a report discussing the results
- Assignment 2 – writing a computer program to solve a given task and a report discussing the results.
- Exam – supervised exam during exam period
- Exam requirement: a minimum of 40% at the exam is required to pass this course

Detailed information for each assessment can be found on Canvas.

Use of generative artificial intelligence (AI)

You can use generative AI tools for open assessments. Restrictions on AI use apply to secure, supervised assessments used to confirm if students have met specific learning outcomes.

Refer to the assessment table above to see if AI is allowed, for assessments in this unit and check Canvas for full instructions on assessment tasks and AI use.

If you use AI, you must always acknowledge it. Misusing AI may lead to a breach of the Academic Integrity Policy.

Visit the Current Students website for more information on AI in assessments, including details on how to acknowledge its use.

Late submission

In accordance with University policy, these penalties apply when written work is submitted after 11:59pm on the due date:

- Deduction of **5%** of the maximum mark for each calendar day after the due date.
- After ten calendar days late, a mark of zero will be awarded.

This unit has an exception to the standard University policy or supplementary information has been provided by the unit coordinator. This information is displayed below:

Weekly homework: No late submissions are allowed. Assignment 1 and Assignment 2: Late submissions are allowed up to 3 days late. A penalty of 5% per day late will apply.

Academic integrity

The University expects students to act ethically and honestly and will treat all allegations of academic integrity breaches seriously.

Our website provides information on academic integrity and the resources available to all students. This includes advice on how to avoid common breaches of academic integrity. Ensure that you have completed the Academic Honesty Education Module (AHEM) which is mandatory for all commencing coursework students

Penalties for serious breaches can significantly impact your studies and your career after graduation. It is important that you speak with your unit coordinator if you need help with completing assessments.

Visit the Current Students website for more information on AI in assessments, including details on how to acknowledge its use.

Learning support

Simple extensions

If you encounter a problem submitting your work on time, you may be able to apply for an extension of five calendar days through a simple extension. The application process will be different depending on the type of assessment and extensions cannot be granted for some assessment types like exams.

Special consideration

If exceptional circumstances mean you can't complete an assessment, you need consideration for a longer period of time, or if you have essential commitments which impact your performance in an assessment, you may be eligible for special consideration or special arrangements.

Special consideration applications will not be affected by a simple extension application.

Using AI responsibly

Co-created with students, AI in Education includes lots of helpful examples of how students use generative AI tools to support their learning. It explains how generative AI works, the different tools available and how to use them responsibly and productively.

Support for students

The Support for Students Policy reflects the University's commitment to supporting students in their academic journey and making the University safe for students. It is important that you read and understand this policy so that you are familiar with the range of support services available to you and understand how to engage with them.

The University uses email as its primary source of communication with students who need support under the Support for Students Policy. Make sure you check your University email regularly and respond to any communications received from the University.

Learning resources and detailed information about weekly assessment and learning activities can be accessed via Canvas. It is essential that you visit your unit of study Canvas site to ensure you are up to date with all of your tasks.

If you are having difficulties completing your studies, or are feeling unsure about your progress, we are here to help. You can access the support services offered by the University at any time:

Support and Services (including health and wellbeing services, financial support and learning support)
Course planning and administration
Meet with an Academic Adviser

Weekly schedule

WK	Topic	Learning activity	Learning outcomes
Week 01	Administrative matters and course overview. Introduction to machine learning and data mining. Data: cleaning, pre-processing and similarity measures.	Lecture (2 hr)	LO1
Week 02	Nearest neighbour. Rule-based algorithms.	Lecture (2 hr)	LO4 LO3 LO2 LO1 LO5
	Nearest neighbour. Rule-based algorithms.	Tutorial (1 hr)	LO4 LO3 LO2 LO1 LO5
Week 03	Linear regression. Logistic regression. Overfitting and regularization.	Lecture (2 hr)	LO4 LO3 LO2 LO1 LO5
	Linear regression. Logistic regression. Overfitting and regularization.	Tutorial (1 hr)	LO4 LO3 LO2 LO1 LO5
Week 04	Probability-based learning. Evaluating machine learning methods.	Lecture (2 hr)	LO4 LO3 LO2 LO1 LO5
	Probability-based learning. Evaluating machine learning methods.	Tutorial (1 hr)	LO4 LO3 LO2 LO1 LO5
Week 05	Decision trees. Ensembles.	Lecture (2 hr)	LO4 LO3 LO2 LO1 LO5
	Decision trees. Ensembles.	Tutorial (1 hr)	LO4 LO3 LO2 LO1 LO5
Week 06	Support vector machines. Kernels. Dimensionality reduction methods.	Lecture (2 hr)	LO4 LO3 LO2 LO1 LO5
	Support vector machines. Kernels. Dimensionality reduction methods.	Tutorial (1 hr)	LO4 LO3 LO2 LO1 LO5
Week 07	Neural networks - perceptrons and backpropagation algorithm.	Lecture (2 hr)	LO4 LO3 LO2 LO1 LO5
	Neural networks - perceptrons and backpropagation algorithm.	Tutorial (1 hr)	LO4 LO3 LO2 LO1 LO5
Week 08	Deep neural networks I – CNN, RNN, and Transformer	Lecture (2 hr)	LO4 LO3 LO2 LO1 LO5
	Deep neural networks I – CNN, RNN, and Transformer	Tutorial (1 hr)	LO4 LO3 LO2 LO1 LO5
Week 09	Deep neural networks II – CNN, RNN, and Transformer	Lecture (2 hr)	LO4 LO3 LO2 LO1 LO5
	Deep neural networks II – CNN, RNN, and Transformer	Tutorial (1 hr)	LO4 LO3 LO2 LO1 LO5
Week 10	Clustering	Lecture (2 hr)	LO4 LO3 LO2 LO1 LO5
	Clustering	Tutorial (1 hr)	LO4 LO3 LO2 LO1 LO5
Week 11	Markov models.	Lecture (2 hr)	LO4 LO3 LO2 LO1 LO5
	Markov models.	Tutorial (1 hr)	LO4 LO3 LO2 LO1 LO5
Week 12	Reinforcement learning.	Lecture (2 hr)	LO4 LO3 LO2 LO1 LO5
	Reinforcement learning.	Tutorial (1 hr)	LO4 LO3 LO2 LO1 LO5
Week 13	Revision and exam review.	Lecture (2 hr)	LO4 LO3 LO2 LO1 LO5
	Revision and exam review.	Tutorial (1 hr)	LO4 LO3 LO2 LO1 LO5

Study commitment

Typically, there is a minimum expectation of 1.5-2 hours of student effort per week per credit point for units of study offered over a full semester. For a 6 credit point unit, this equates to roughly 120-150 hours of student effort in total.

Required readings

Textbooks:

1. Ian H. Witten, Eibe Frank, Mark Hall and Christopher J. Pal (2017). *Data Mining - Practical Machine Learning Tools and Techniques*, 4th edition, Morgan Kaufmann.

2. Pang-Ning Tan, Michael Steinbach, Anuj Karpathe and Vipin Kumar (2019). *Introduction to Data Mining*, 2nd edition, Pearson.

Books for the practical part using Python:

1. Andreas C. Mueller and Sarah Guido (2016). *Introduction to Machine Learning with Python: A Guide for Data Scientists*, O'Reilly.

2. Aurelien Geron (2022). *Hands-On Machine Learning with Scikit-Learn, Keras and TensorFlow*, Second edition, O'Reilly.

Learning outcomes

Learning outcomes are what students know, understand and are able to do on completion of a unit of study. They are aligned with the University's graduate qualities and are assessed as part of the curriculum.

Outcomes | Graduate qualities

At the completion of this unit, you should be able to:

LO1. LO1: Explain the principles, assumptions, strengths, weaknesses, and applicability of machine learning algorithms for classification, regression, clustering, and reinforcement learning tasks.
LO2. LO2: Apply machine learning software tools and libraries to prepare data, train models, tune parameters, and generate predictions.
LO3. LO3: Analyse and compare machine learning algorithms using appropriate evaluation metrics and problem-specific criteria.
LO4. LO4: Design, implement, and evaluate a complete machine learning solution, justifying algorithm choices and interpreting outcomes.
LO5. LO5: Communicate machine learning methods, results, and interpretations clearly in verbal and written forms using suitable visualisations and evidence.

Responding to student feedback

This section outlines changes made to this unit following staff and student reviews.

Unit learning outcomes were revised for S2, 2026, and redistributed across assessments.

Disclaimer

Important: the University of Sydney regularly reviews units of study and reserves the right to change the units of study available annually. To stay up to date on available study options, including unit of study details and availability, refer to the relevant handbook.

The information included on this page is correct at 31 Jul 2026.

To help you understand common terms that we use at the University, we offer an online glossary.

We recognise and pay respect to the Elders and communities - past, present, and emerging - of the lands that the University of Sydney's campuses stand on. For thousands of years they have shared and exchanged knowledges across innumerable generations for the benefit of all.

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